

Light-Phase Substrate Theory (LPST): A Relational Phase-Interference Model of Emergent Space, Time, Matter, and Gravity

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Abstract

We introduce Light-Phase Substrate Theory (LPST), a framework in which light-like phase interference constitutes the sole fundamental substrate of physical reality. Space, time, matter, energy, and gravity are not primitive entities but emergent structures derived from correlations, topological defects, and phase-consistency constraints within a universal complex field. LPST replaces background spacetime dynamics with a global interference-closure principle and derives geometry from correlation kernels rather than postulated metrics. Matter arises as stable topological solitons—knots, vortices, and braids—of the phase and internal polarization manifolds of the substrate. Gravitational phenomena emerge as self-induced refraction due to local modulation of phase stiffness. The theory naturally reproduces relativistic kinematics, quantum interference, and mass-energy equivalence while suggesting falsifiable deviations in extreme regimes.

1. Introduction and Motivation

The two pillars of twentieth-century physics—quantum mechanics and general relativity—have proven spectacularly successful within their respective domains, yet they remain fundamentally incompatible at the structural level. General relativity describes gravity as the curvature of a smooth, classical spacetime manifold

governed by the Einstein field equations, while quantum mechanics operates on a fixed background spacetime and describes matter and radiation in terms of probability amplitudes, superposition, and entanglement. Attempts to quantize gravity by applying the perturbative methods that succeeded for electromagnetism and the nuclear forces encounter non-renormalizable ultraviolet divergences, signaling that the union of these frameworks demands a conceptual shift rather than a mere technical refinement.

Over the past three decades, a convergent body of evidence from quantum information theory, black hole thermodynamics, and the holographic principle has suggested that spacetime itself may not be fundamental. The Bekenstein-Hawking entropy formula $S = A/(4G)$, relating the entropy of a black hole to the area of its event horizon rather than its volume, was an early indication that gravitational degrees of freedom are holographic in nature [1,2]. The AdS/CFT correspondence, proposed by Maldacena in 1997 [3], provided a concrete realization of this idea: the geometry of an anti-de Sitter bulk spacetime is encoded in the correlators of a conformal field theory living on its boundary. The Ryu-Takayanagi formula [4] further sharpened this picture by identifying the entanglement entropy of boundary subregions with the areas of minimal surfaces in the bulk, directly linking quantum entanglement to spatial geometry.

Jacobson's landmark 1995 derivation of the Einstein equations from thermodynamic considerations [5] demonstrated that the equations governing spacetime curvature can be viewed as an equation of state—a macroscopic relation emerging from underlying microscopic degrees of freedom, much as the ideal gas law emerges from molecular kinetics. Verlinde's entropic gravity proposal [6] extended this thermodynamic reasoning, arguing that gravitational attraction is an entropic force arising from changes in information associated with material bodies. Padmanabhan's extensive program [7,8] further developed the thermodynamic perspective, showing that the gravitational field equations in a wide class of theories can be rewritten as thermodynamic identities on null surfaces. The ER=EPR conjecture of Maldacena and Susskind [9] and Van Raamsdonk's work on entanglement and spacetime connectivity [10] have deepened the case that quantum correlations are the fabric from which geometry is woven.

In the present work, we observe three key features of known physics that, taken together, motivate a radical reformulation:

(1) Interference precedes geometry. Quantum amplitudes and phase relationships appear to be more fundamental than spatial localization. The double-slit experiment, quantum entanglement, and the path integral formulation all indicate that physics is first and foremost about the coherent superposition of complex amplitudes. Spatial structure is inferred from the pattern of interference, not the other way around.

(2) Geometry tracks information flow. Distance, elapsed time, and curvature correlate with signal propagation and correlation decay in every known physical context. The causal structure of spacetime is defined by light-cone boundaries, the metric can be reconstructed from the propagation of light signals (radar coordinates), and the holographic principle ties geometry to information content. This suggests that geometry *is* correlation structure.

(3) Matter behaves as structured light. The mass-energy equivalence $E = mc^2$ tells us that mass is a form of energy. Gravitational lensing demonstrates that light and matter interact on equal footing with the geometry they jointly define. Pair production and annihilation show that matter can be created from and destroyed into pure radiation. The Compton wavelength and de Broglie relations assign wave properties to all matter. These facts suggest that the distinction between light and matter is one of organization rather than substance.

LPST formalizes these observations by postulating that *light-like phase interference is the sole primitive entity*. There is no background spacetime, no fundamental gravitational field, and no matter fields distinct from the substrate. All of these emerge from the topology, correlations, and self-consistency constraints of a single universal complex field. This approach shares the background-independence aspiration of loop quantum gravity [11,12] and the unification ambition of string theory [13], but differs from both in several critical respects. Unlike loop quantum gravity, LPST does not discretize geometry at the fundamental level; the substrate is a continuous field, and discreteness (quantization) emerges from topological constraints. Unlike string theory, LPST does not embed its fundamental entity in a pre-existing spacetime; rather, spacetime emerges from the substrate's correlation

structure. Unlike causal set theory [14], which posits a discrete partial order as fundamental, LPST derives causal ordering from the phase-update structure of a continuous field.

The structure of this paper is as follows. Section 2 provides historical context and surveys prior work. Section 3 establishes notation and conventions. Section 4 states the foundational postulates. Sections 5–8 develop the mathematical structure of the substrate and its dynamics. Sections 9–12 show how spacetime geometry, causal structure, time, and Lorentz invariance emerge. Sections 13–18 treat matter as topological defects and derive gauge interactions. Sections 19–21 develop the gravitational sector. Sections 22–24 recover quantum mechanics and thermodynamics. Sections 25–27 address cosmology, black holes, and dark sector phenomenology. Sections 28–31 compare LPST to other approaches, discuss computational methods, experimental predictions, and open problems. Section 32 concludes. Appendices provide detailed derivations.

2. Historical Context and Prior Work

LPST draws upon a rich intellectual lineage spanning nearly a century of theoretical physics. In this section we survey the key antecedents and clarify how LPST both builds upon and departs from each.

2.1 Wheeler’s “It from Bit” and Geometrodynamics

John Archibald Wheeler’s program of geometrodynamics [15] sought to describe all of physics in terms of the geometry and topology of spacetime. His later aphorism “it from bit” [16] proposed that every physical quantity derives its meaning from information-theoretic yes-or-no questions. LPST inherits Wheeler’s conviction that information and topology are primary, but replaces his spacetime foam with the phase-interference substrate: geometry does not fluctuate at the Planck scale; rather, it emerges from correlations in a field that is not itself geometric.

2.2 Penrose’s Twistor Theory and Spin Networks

Penrose’s twistor program [17] reformulates spacetime physics in terms of null structures—the trajectories of massless particles. Twistors are objects that naturally encode the conformal geometry of Minkowski space and are fundamentally light-like. LPST shares this emphasis on null structures: the substrate is “light-like” in that its fundamental excitations propagate at the invariant speed, and geometry is secondary to the null-ray structure. Penrose’s spin networks [18], later adopted and developed in loop quantum gravity, describe discrete quantum geometry. LPST’s topological defects play an analogous role to spin-network nodes and edges but arise within a continuous field rather than as a discrete combinatorial structure.

2.3 Thermodynamic and Entropic Gravity

Jacobson’s 1995 derivation [5] showed that the Einstein equations follow from the demand that the Clausius relation $\delta Q = T dS$ hold on all local Rindler horizons, with entropy proportional to horizon area and temperature given by the Unruh formula. This was a profound hint that gravity is emergent. Verlinde [6] extended this to argue that gravitational force is entropic, arising from changes in information content. Padmanabhan [7,8] demonstrated the thermodynamic identity structure of gravitational field equations in Lanczos-Lovelock theories. LPST provides a concrete microscopic substrate whose coarse-grained description reproduces these thermodynamic relations: the phase stiffness of the substrate determines the “temperature” of the emergent geometry, and the entropy is the coarse-grained count of phase configurations.

2.4 Holography, AdS/CFT, and ER=EPR

The AdS/CFT correspondence [3] provides a complete dictionary between boundary quantum field theory correlators and bulk gravitational physics. The Ryu-Takayanagi formula [4] identifies boundary entanglement entropy with bulk minimal surface areas. Van Raamsdonk [10] argued compellingly that entanglement is the “glue” holding spacetime together: disentangling boundary regions disconnects the bulk geometry. The ER=EPR conjecture [9] proposed that entangled particles are connected by non-traversable wormholes. LPST’s correlation-to-geometry dictionary

(Section 9) is the analogue of these holographic relations, but operates in a more general setting: no asymptotically AdS boundary conditions are required, and the map is from substrate correlations to emergent geometry everywhere, not only at asymptotic infinity.

2.5 Causal Dynamical Triangulations and Discrete Approaches

Causal dynamical triangulations (CDT) [19] construct spacetime from a path integral over piecewise-flat simplicial geometries with a built-in causal structure. The Regge calculus [20] discretizes general relativity on simplicial lattices. These approaches have demonstrated that sensible semiclassical geometries can emerge from sums over discrete building blocks. LPST takes a complementary route: instead of discretizing geometry and summing, it starts with a continuous field and shows that effective geometry emerges from correlations. The two approaches may ultimately be related via a coarse-graining map.

2.6 Optical Geometry and Analogue Gravity

Gordon [21] showed in 1923 that light propagation in a moving dielectric medium is equivalent to propagation in a curved effective spacetime. Leonhardt and Piwnicki [22] developed this into the analogue gravity program, and Unruh [23] demonstrated that sonic horizons in supersonic fluid flows emit thermal phonon radiation analogous to Hawking radiation. These results demonstrate that effective spacetimes can emerge from underlying non-gravitational physics. LPST elevates this analogy to a fundamental principle: gravity *is* refraction in the phase substrate, not merely analogous to it.

3. Notation and Conventions

We adopt the following conventions throughout this paper unless explicitly stated otherwise.

3.1 Units and Indices

Natural units. We set $c = \hbar = 1$ except in sections where the explicit dependence on these constants is pedagogically useful. Newton’s constant G is retained as a derived quantity expressed in terms of substrate parameters.

Metric signature. When the emergent metric is established, we use the “mostly plus” signature $(-, +, +, +)$.

Index conventions. Greek indices $\mu, \nu, \rho, \sigma = 0, 1, 2, 3$ label emergent spacetime directions. Latin indices a, b, c from the beginning of the alphabet label coordinates on the internal polarization manifold M_{int} . Latin indices i, j, k from the middle of the alphabet label spatial directions 1, 2, 3.

3.2 The Substrate Field

The universal complex field is denoted:

$$L(x) = A(x) e^{i\varphi(x)} \tag{1}$$

where $A(x) \geq 0$ is the real amplitude and $\varphi(x) \in [0, 2\pi)$ is the phase. Here x is a label in an abstract label space—not a point in a pre-existing spacetime. The internal polarization state is a unit vector:

$$\mathbf{u}(x) \in S^2 \subset \mathbb{R}^3, \quad |\mathbf{u}|^2 = 1 \tag{2}$$

More generally, $\mathbf{u}$ may take values in an internal manifold M_{int} whose isometry group determines the emergent gauge symmetry.

3.3 Key Parameters

Symbol	Name	Description
α	Phase stiffness coefficient	Energy cost per unit phase gradient squared
β	Polarization coupling	Coupling of polarization gradients to amplitude
γ	Topological stabilizer coupling	Coefficient of the higher-order topological term
λ	Potential coupling	Strength of the Mexican-

Symbol	Name	Description
		hat potential
A_0	Vacuum amplitude	Equilibrium value of $ L $ in the ground state
ξ	Correlation length	Characteristic decay scale of $C(x,y)$ in vacuum
$\kappa(x)$	Local phase stiffness	$\kappa(x) = \alpha L(x) ^2$

3.4 Derivatives and Differential Operators

Partial derivatives are denoted ∂_μ . When the emergent metric $g_{\mu\nu}$ has been established, the covariant derivative is ∇_μ , with the Levi-Civita connection determined by $g_{\mu\nu}$. The totally antisymmetric tensor is $\varepsilon^{\mu\nu\rho\sigma}$ with $\varepsilon^{0123} = +1/\sqrt{-g}$. The correlation kernel is denoted $C(x,y) = \langle L(x) L^*(y) \rangle$.

4. Foundational Postulates

LPST rests on four postulates that collectively define the theory. We state each formally and provide extensive commentary.

4.1 Postulate 1 — Single Substrate

There exists a universal complex field $L = A e^{i\varphi}$, equipped with an internal polarization state $\mathbf{u}$, which constitutes the sole fundamental physical substrate. All physical entities—space, time, matter, energy, forces—are emergent structures within L .

This postulate represents a radical form of ontological monism. There is exactly one kind of “stuff” in the universe: the substrate field L and its internal structure. This is more parsimonious than the Standard Model, which posits dozens of fundamental fields, and more parsimonious than string theory, which postulates a one-dimensional string but embeds it in a ten- or eleven-dimensional background spacetime. In LPST, L is not embedded in anything. The field L is the totality of physical reality.

The closest analogue in existing physics is the universal wavefunction Ψ of Everettian quantum mechanics [24], which also claims to describe all of reality as a

single mathematical object. The key difference is that Ψ is defined on a pre-existing configuration space, whereas L is self-referential: it generates its own effective spacetime through its correlations. Another analogue is the string field of string field theory, which contains all string excitations; however, string field theory still requires a background spacetime for its formulation.

The optional internal polarization state $\mathbf{u}$ is not an additional postulate but a necessary structure for accommodating the observed richness of physics: without internal degrees of freedom, the field $L = A e^{i\varphi}$ supports only Abelian (U(1)) topological structures, which are insufficient to reproduce the non-Abelian gauge symmetries of the Standard Model. The introduction of $\mathbf{u}$ on an internal manifold M_{int} provides the requisite topological complexity.

4.2 Postulate 2 — No Background Spacetime

Space and time are not fundamental coordinates. They emerge as effective parameters describing, respectively, the correlation structure and the phase-update ordering within L .

This postulate addresses the “problem of time” in quantum gravity [25] and the demand for background independence [11]. In canonical quantum gravity, the Wheeler-DeWitt equation $H\Psi = 0$ contains no time parameter, and the recovery of dynamical evolution is notoriously problematic. In LPST, this problem is dissolved rather than solved: time is not a parameter in the fundamental description but emerges from the internal dynamics of the substrate, as we develop in Section 11.

Background independence means that the fundamental formulation of the theory makes no reference to a pre-existing metric structure. The field L is defined on an abstract label space, and all geometric notions (distance, angle, volume, curvature) are derived from the correlations and topology of L . This is a stronger form of background independence than that achieved in loop quantum gravity, where the Hilbert space of spin-network states is background-independent but the framework still relies on a differentiable manifold as an intermediate structure.

4.3 Postulate 3 — Phase Consistency Principle

Allowed physical histories are those that minimize global phase inconsistency. Dynamics arise from interference closure rather than time-first evolution.

We define the phase inconsistency functional as:

$$I[\varphi] = \oint (\nabla\varphi \cdot d\mathbf{l} - 2\pi n) \text{ summed over all closed loops (3)}$$

where n is the integer winding number appropriate to each loop. Physical configurations satisfy $\delta I = 0$ for all variations that preserve topological charges. This principle replaces the role of a Hamiltonian or Lagrangian at the most fundamental level: dynamics is not generated by a time-translation operator but by the requirement of global self-consistency of the phase structure.

The phase consistency principle is the LPST analogue of the principle of stationary action, but it is more primitive. The stationary-action principle presupposes a time coordinate along which the action is integrated; the phase consistency principle does not. It is a global constraint on the allowed configurations of L , analogous to the requirement that the wavefunction be single-valued in quantum mechanics. In the regime where an effective spacetime has emerged (Sections 9–11), the phase consistency principle reduces to the stationary-action principle for the substrate action (Section 6).

4.4 Postulate 4 — Geometry from Correlation

Metric structure, causal cones, and curvature are derived from the correlation kernel $C(x,y) = \langle L(x) L^(y) \rangle$ of the substrate, not imposed a priori.*

This postulate provides the bridge from the abstract substrate to the familiar geometric arena of physics. The correlation kernel $C(x,y)$ is the two-point function of L : it encodes how similar the substrate configuration is at two label-space points x and y . From $C(x,y)$, we extract an effective metric $g_{\mu\nu}$ (Section 9), causal structure (Section 10), and curvature. This is analogous in spirit to the Ryu-Takayanagi prescription [4], which extracts geometric data from entanglement entropy, and to the reconstruction of bulk geometry from boundary correlators in AdS/CFT [3]. However, LPST's correlation-to-geometry map is more general: it does not require

an asymptotic boundary or anti-de Sitter geometry, and it operates at the level of the substrate’s own correlations rather than those of a dual field theory.

5. The Universal Complex Field

In this section we develop the mathematical characterization of the substrate field L in detail, discuss its physical interpretation, and establish the framework for perturbative analysis around the vacuum.

5.1 Mathematical Structure

Formally, L is a section of a complex line bundle over the abstract label space Σ , which we take to be a four-dimensional smooth manifold equipped only with a differentiable structure (no metric). More precisely, the full substrate field is a section of a fiber bundle with typical fiber $\mathbb{C} \times M_{\text{int}}$, where M_{int} is a compact manifold encoding the internal polarization degrees of freedom. For the minimal model discussed in this paper, $M_{\text{int}} = S^2$ (the two-sphere), so the full configuration space at each label-space point is $\mathbb{C} \times S^2 \cong \mathbb{R}_{\geq 0} \times S^1 \times S^2$, after decomposing L into amplitude and phase.

The field $L(x)$ encodes all information about the state of reality at each label $x \in \Sigma$. We emphasize that x is a label, not a “point in space”—spatial notions have not yet been defined. The labels serve only to distinguish different degrees of freedom of the substrate, much as indices label components of a vector.

5.2 Physical Interpretation

The amplitude $A(x) = |L(x)|$ represents the intensity or “presence” of the substrate at label x . Regions where A is large are strongly present; regions where $A \rightarrow 0$ represent the cores of topological defects where the substrate description requires topological data (winding numbers, linking numbers) rather than smooth field values. The phase $\varphi(x)$ represents the oscillatory, wave-like degree of freedom. Phase gradients carry energy and momentum; phase differences between labels encode all dynamical information. The polarization $\mathbf{u}(x)$ adds internal structure that enables the substrate

to support non-Abelian topological defects (skyrmions, hopfions) and thereby to accommodate non-Abelian gauge symmetries and fermionic statistics.

5.3 The Vacuum State

The vacuum (ground state) of the substrate is the maximally coherent configuration:

$$L_{\text{vac}}(\mathbf{x}) = A_0 e^{i\varphi_0}, \quad \mathbf{u}_{\text{vac}}(\mathbf{x}) = \mathbf{u}_0 = \text{const.} \quad (4)$$

This is the state of maximal correlation: $C(\mathbf{x}, \mathbf{y}) = A_0^2$ for all $\mathbf{x}, \mathbf{y}$. All phase gradients vanish, all polarization gradients vanish, and the potential energy is minimized. The vacuum is homogeneous and isotropic in the label space, and it is from perturbations of this vacuum that all physics emerges.

5.4 Gauge Redundancy

The overall phase φ_0 and the overall orientation $\mathbf{u}_0$ of the vacuum are unobservable. Only phase *differences* $\Delta\varphi(\mathbf{x}, \mathbf{y}) = \varphi(\mathbf{x}) - \varphi(\mathbf{y})$ and relative orientations of $\mathbf{u}$ are physical. This gives rise to global gauge symmetries: U(1) phase rotations and SO(3) rotations of $\mathbf{u}$. When these symmetries are promoted to local symmetries via the topological structure of defects, they become the gauge symmetries of the emergent theory (Section 16).

6. Substrate Action and Lagrangian Density

The dynamics of the substrate are governed by the action:

$$S = \int d^4x \sqrt{(-g)} \left[\alpha g^{\mu\nu} \partial_\mu L^* \partial_\nu L - V(|L|) + \beta |L|^2 g^{\mu\nu} \partial_\mu \mathbf{u} \cdot \partial_\nu \mathbf{u} + \gamma K(\varphi, \mathbf{u}) \right] \quad (5)$$

We now discuss each term in detail.

6.1 The Kinetic (Phase Stiffness) Term

The first term, $\alpha g^{\mu\nu} \partial_\mu L^* \partial_\nu L$, is the kinetic energy of the substrate field. The parameter $\alpha > 0$ is the phase stiffness coefficient: it controls the energy cost associated with spatial and temporal gradients of L . When L is decomposed into

amplitude and phase (Section 7), this term generates both the amplitude gradient energy (controlling defect core structure) and the phase gradient energy (which, in the low-energy sector, becomes kinetic energy, momentum, and mass). The coefficient α sets the overall energy scale of the theory and, together with A_0 , determines Newton’s gravitational constant (Section 20).

6.2 The Potential $V(|L|)$

The potential takes the Mexican-hat (double-well) form:

$$V(|L|) = \lambda(|L|^2 - A_0^2)^2 \quad (6)$$

This potential has a minimum at $|L| = A_0$ and a local maximum at $|L| = 0$. It stabilizes the vacuum amplitude: perturbations that attempt to change A away from A_0 are energetically penalized. The parameter λ controls the stiffness of the amplitude mode (the “Higgs-like” radial excitation of L). For $\lambda \gg \alpha/\xi^2$, the amplitude mode is massive and can be integrated out at energies below its mass, leaving only the phase and polarization degrees of freedom—the Goldstone sector. The spontaneous breaking of the global $U(1)$ symmetry $L \rightarrow L e^{i\theta}$ by the vacuum $A = A_0 \neq 0$ is the substrate analogue of the Higgs mechanism, and the massless Goldstone boson of this broken symmetry is the photon-like excitation of the phase field.

6.3 The Polarization Kinetic Term

The term $\beta |L|^2 g^{\mu\nu} \partial_\mu \mathbf{u} \cdot \partial_\nu \mathbf{u}$ is a nonlinear sigma model for the polarization field $\mathbf{u}$, coupled to the substrate amplitude $|L|^2$. The coupling to $|L|^2$ ensures that polarization gradients are energetically significant only where the substrate is “present” ($A \neq 0$). This term supports skyrmion-type topological solitons in the polarization field: stable, localized configurations of $\mathbf{u}$ that wrap the target sphere S^2 a non-trivial number of times. These skyrmions are the LPST candidates for baryonic matter (Section 14).

6.4 The Topological Stabilizer $K(\varphi, \mathbf{u})$

The term $\gamma K(\varphi, \mathbf{u})$ is a higher-order topological term whose role is to prevent topological defects from collapsing to zero size. Without it, Derrick’s theorem [26]

implies that static, finite-energy soliton solutions do not exist in three spatial dimensions for the standard kinetic terms alone. The topological stabilizer K may take several forms:

- **Skyrme term:** $K = |[\partial_\mu \mathbf{u} \times \partial_\nu \mathbf{u}]|^2$, a quartic derivative term that provides a positive energy contribution scaling as $1/R$ (where R is the soliton size), balancing the kinetic energy contribution that scales as R [27].
- **Hopf invariant density:** $K = \mathbf{F} \cdot \mathbf{A}$, where $\mathbf{F} = d\mathbf{A}$ is the field strength of a connection derived from $\mathbf{u}$. This term is topological and stabilizes hopfion (knotted soliton) configurations [28].
- **BF-type coupling:** $K = B \wedge F$, coupling the phase curvature to the polarization curvature. This term has roots in topological field theory [29].

The coefficient $\gamma > 0$ sets the size scale of the stabilized defects and, together with the other parameters, determines the mass spectrum of topological solitons.

6.5 Self-Referential Nature of the Action

A crucial subtlety: the metric $g_{\mu\nu}$ appearing in the action (5) is *not* an independent fundamental field. It is determined self-consistently from the correlations of L via the prescription of Section 9. The action is therefore implicit: the dynamics of L depend on the metric, which depends on the correlations of L , which depend on the dynamics. This self-referential structure is the mathematical expression of the bootstrapping nature of LPST—the universe, in this framework, is a self-consistent interference pattern. In practice, one solves the theory iteratively: starting from a trial metric, computing the field dynamics and correlations, extracting the implied metric, and iterating until self-consistency is achieved.

7. Phase-Amplitude Decomposition

Substituting $L = A e^{i\varphi}$ into the kinetic term, we compute:

$$\partial_\mu L^* \partial^\mu L = \partial_\mu (A e^{-i\varphi}) \partial^\mu (A e^{i\varphi}) \quad (7)$$

Expanding the derivatives:

$$\partial_\mu(A e^{-i\varphi}) = (\partial_\mu A) e^{-i\varphi} - iA (\partial_\mu \varphi) e^{-i\varphi} \quad (8)$$

$$\partial_\mu(A e^{i\varphi}) = (\partial_\mu A) e^{i\varphi} + iA (\partial_\mu \varphi) e^{i\varphi} \quad (9)$$

Multiplying and contracting:

$$\partial_\mu L^* \partial^\mu L = (\partial_\mu A)(\partial^\mu A) + A^2(\partial_\mu \varphi)(\partial^\mu \varphi) \quad (10)$$

where the cross terms cancel due to the opposite signs of the $\pm i$ contributions. This decomposition has a transparent physical interpretation:

Amplitude gradient energy: The term $(\partial_\mu A)(\partial^\mu A)$ represents the energy cost of spatial and temporal variations in the substrate intensity. This term is important in the cores of topological defects, where A varies rapidly from its vacuum value A_0 down to zero. In the low-energy, long-wavelength regime where $A \approx A_0 = \text{const}$, this term is negligible.

Phase gradient energy: The term $A^2(\partial_\mu \varphi)(\partial^\mu \varphi)$ is the dominant contribution in the low-energy sector. This term represents the kinetic energy of phase waves and, for topological defects, the trapped phase energy that constitutes their mass. The factor A^2 means that the effective stiffness of the phase field is modulated by the local amplitude—this is the origin of self-induced refraction and emergent gravity (Section 19).

The full Lagrangian density in the decomposed form reads:

$$\mathcal{L} = \alpha[(\partial_\mu A)^2 + A^2(\partial_\mu \varphi)^2] - \lambda(A^2 - A_0^2)^2 + \beta A^2 (\partial_\mu \mathbf{u})^2 + \gamma K(\varphi, \mathbf{u}) \quad (11)$$

In the vacuum sector ($A = A_0 = \text{const}$), this reduces to:

$$\mathcal{L}_{\text{vac}} = \alpha A_0^2 (\partial_\mu \varphi)^2 + \beta A_0^2 (\partial_\mu \mathbf{u})^2 + \gamma K(\varphi, \mathbf{u}) \quad (12)$$

This is a coupled nonlinear sigma model for the phase φ and polarization $\mathbf{u}$, with the Skyrme-type stabilizer term. In this regime, all familiar physics operates. The energy-momentum tensor of the substrate is obtained by varying the action with respect to $g_{\mu\nu}$:

$$T_{\mu\nu} = \alpha[(\partial_\mu A)(\partial_\nu A) + A^2(\partial_\mu \varphi)(\partial_\nu \varphi)] + \beta A^2(\partial_\mu \mathbf{u} \cdot \partial_\nu \mathbf{u}) - g_{\mu\nu} \mathcal{L} + \gamma T_{\mu\nu}^{(K)} \quad (13)$$

where $T_{\mu\nu}^{(K)}$ is the energy-momentum contribution from the topological stabilizer. This energy-momentum tensor, evaluated for topological defect configurations, yields the source terms for the emergent gravitational field.

8. Variational Principles and Field Equations

Applying the Euler-Lagrange equations to the action (5) yields the coupled field equations for the three substrate degrees of freedom.

8.1 Amplitude Equation

Varying the action with respect to A yields:

$$\alpha \nabla^2 A - \alpha A (\partial_\mu \varphi)^2 - 2\lambda A(A^2 - A_0^2) - \beta A (\partial_\mu \mathbf{u})^2 = 0 \quad (14)$$

where $\nabla^2 = g^{\mu\nu} \nabla_\mu \nabla_\nu$ is the covariant d'Alembertian. This equation describes the balance between the tendency of A to remain at its vacuum value A_0 (enforced by the potential) and the “centrifugal” effects of phase and polarization gradients (which tend to suppress A , creating defect cores). The term $-\alpha A (\partial_\mu \varphi)^2$ is particularly significant: it shows that regions of large phase gradient experience a reduction of A , which in turn modifies the local phase stiffness $\kappa(x) = \alpha A^2$ —the mechanism underlying emergent gravity.

8.2 Phase Equation

Varying with respect to φ :

$$\nabla_\mu (\alpha A^2 \nabla^\mu \varphi) + \gamma \partial K / \partial \varphi = 0 \quad (15)$$

This is a conservation law: the quantity $j^\mu = \alpha A^2 \nabla^\mu \varphi$ is a conserved current (up to the topological source term from K). In the vacuum ($A = A_0$, K negligible), this reduces to the massless wave equation:

$$\nabla^2 \varphi = 0 \quad (16)$$

confirming that small phase excitations propagate at the speed of light (defined as the characteristic speed of the emergent metric). This is the LPST realization of electromagnetic wave propagation (Section 17).

8.3 Polarization Equation

Varying with respect to $\mathbf{u}$, subject to the constraint $|\mathbf{u}|^2 = 1$:

$$\beta \nabla_\mu (A^2 \nabla^\mu \mathbf{u}) + \beta A^2 (\nabla_\mu \mathbf{u} \cdot \nabla^\mu \mathbf{u}) \mathbf{u} + \gamma \partial K / \partial \mathbf{u} = 0 \quad (17)$$

The second term enforces the constraint $|\mathbf{u}| = 1$ via a Lagrange multiplier absorbed into the equations of motion. This is the equation of motion of a nonlinear sigma model with target space S^2 , coupled to the amplitude field. It admits skyrmion solutions: localized configurations where $\mathbf{u}$ wraps around S^2 one or more times.

8.4 The Coupled System and the Phase Consistency Selection Rule

Equations (14)–(17) form a coupled, nonlinear system of partial differential equations. Not all solutions of this system are physical. Postulate 3 (the phase consistency principle) acts as an additional selection rule: only those solutions for which the global phase structure is self-consistent—meaning that the phase inconsistency functional $I[\varphi]$ (Eq. 3) is minimized—are admitted as physical configurations. This selection rule eliminates pathological solutions (e.g., configurations with unphysical branch cuts or inconsistent winding numbers) and, in the path-integral formulation (Section 22), determines the measure over field configurations.

9. Emergence of Metric Geometry from Correlations

This section develops the central mechanism of LPST: the derivation of an effective spacetime metric from the correlation structure of the substrate.

9.1 The Correlation Kernel

We define the two-point correlation kernel as the vacuum expectation value (or, in the absence of a pre-defined notion of quantum state, the statistical average over the ensemble of phase-consistent configurations):

$$C(x,y) = \langle L(x) L^*(y) \rangle \quad (18)$$

In the vacuum state (4), $L(x) = A_0 e^{i\phi_0}$ is spatially uniform, and $C(x,y) = A_0^2$ for all x, y . When the substrate is perturbed, phase fluctuations cause $C(x,y)$ to decay with increasing “separation” between x and y (where “separation” is initially defined only in the label space and acquires geometric meaning through the construction below).

9.2 Emergent Distance

We define the emergent squared distance between two label-space points as:

$$d^2(x,y) = -\ell^2 \ln(|C(x,y)| / |C(x,x)|) \quad (19)$$

where ℓ is a length scale set by the correlation length ξ . This definition has the following properties: (i) $d(x,x) = 0$ since $C(x,x)/C(x,x) = 1$ and $\ln(1) = 0$; (ii) $d(x,y) \geq 0$ since correlations decay, so $|C(x,y)| \leq |C(x,x)|$; (iii) $d(x,y) = d(y,x)$ if C is Hermitian; (iv) the triangle inequality is satisfied for a broad class of correlation kernels, including all kernels of the form $C(x,y) = A_0^2 \exp(-f(x,y))$ with f a metric function.

9.3 Derivation of the Effective Metric

For nearby points $y = x + \delta x$, we Taylor-expand the correlation kernel. Consider a vacuum-sector kernel of the Gaussian form:

$$C(x,y) = A_0^2 \exp(-|x - y|^2 / (2\xi^2)) \quad (20)$$

Substituting into (19):

$$d^2(x,y) = -\ell^2 \ln(\exp(-|x - y|^2 / (2\xi^2))) = (\ell^2/(2\xi^2)) |x - y|^2 \quad (21)$$

Setting $\ell^2/(2\xi^2) = 1$ (a choice of units), we recover $d^2(x,y) = |x - y|^2$, i.e., flat Euclidean distance. For the general case, the expansion of $C(x, x + \delta x)$ to second order in δx gives:

$$\ln|C(x, x+\delta x)| \approx \ln|C(x, x)| + \partial_\mu \ln|C| \delta x^\mu + \frac{1}{2} \partial_\mu \partial_\nu \ln|C| \delta x^\mu \delta x^\nu + \dots \quad (22)$$

The first-order term vanishes at $y = x$ by the symmetry $C(x, y) = C(y, x)^*$, so:

$$d^2(x, x+\delta x) \approx -\ell^2 \cdot \frac{1}{2} \partial_\mu \partial_\nu \ln|C(x, y)| \Big|_{y=x} \delta x^\mu \delta x^\nu \quad (23)$$

Comparing with the standard metric line element $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$, we identify the emergent metric:

$$g_{\mu\nu}(x) = -(\ell^2/2) \partial_\mu \partial_\nu \ln|C(x, y)| \Big|_{y=x} \quad (24)$$

This is the fundamental equation of emergent geometry in LPST. It expresses the metric—the object that encodes all of spatial distance, temporal duration, and causal structure—as a derived quantity computed from the correlation kernel of the substrate.

9.4 How Defects Curve Geometry

In the presence of a topological defect at label-space point x_0 , the amplitude A is suppressed near x_0 , and the phase φ has a non-trivial winding structure. Both effects modify $C(x, y)$ relative to its vacuum form. Specifically, near the defect the correlation kernel takes the form:

$$C(x, y) \approx A_0^2 \exp(-|x-y|^2/(2\xi^2)) \cdot f(x, x_0) f^*(y, x_0) \quad (25)$$

where f encodes the distortion due to the defect. The extra terms in $\ln|C|$ modify the second derivatives in Eq. (24), producing a non-flat metric—curvature. The defect curves the emergent geometry precisely because it modifies the correlation structure. This is how matter (a topological defect) generates gravity (curvature of the emergent metric) in LPST.

9.5 Comparison to Holographic Prescriptions

The correlation-to-metric map (24) is analogous to the bulk metric reconstruction from boundary correlators in AdS/CFT. In that context, the boundary two-point function $\langle O(x) O(y) \rangle \sim 1/|x-y|^{2\Delta}$ encodes the conformal dimension Δ and, through the HKLL reconstruction [30], the bulk geometry. LPST's prescription is more direct: it operates within the substrate itself rather than between a boundary and a bulk, and

it does not require conformal symmetry or an AdS background. Nevertheless, the structural parallel—correlations encode geometry—suggests a deep connection between LPST and holographic duality that deserves further investigation.

10. Causal Structure and Light Cones

With the emergent metric $g_{\mu\nu}$ in hand, the causal structure of the emergent spacetime is determined by the null cone: the set of directions δx^μ satisfying $g_{\mu\nu} \delta x^\mu \delta x^\nu = 0$. We now show that this causal structure has a natural interpretation in terms of the substrate.

10.1 Null Geodesics as Paths of Maximal Correlation

In the emergent geometry, null geodesics are the paths along which light propagates. In the substrate picture, light is a propagating phase disturbance (Section 17). The null condition $g_{\mu\nu} dx^\mu dx^\nu = 0$ corresponds, via Eq. (24), to the condition that the correlation kernel $C(x,y)$ does not decay along the null direction. More precisely, null geodesics are paths γ in label space along which the rate of correlation decay per unit affine parameter is zero: $d/d\lambda[\ln|C(x(\lambda), x_{\text{ref}})|] = 0$ for a fixed reference point x_{ref} at lowest order. These are the paths of maximal phase correlation—the substrate analogue of light rays.

10.2 Causal Ordering from Phase Updates

We define the causal relation: label x is in the causal past of label y , written $x < y$, if there exists a chain of maximally correlated (null or timelike) substrate labels connecting x to y along which the phase is monotonically advancing. This provides a partial ordering on the label space that, in the vacuum, reproduces the standard Minkowski causal structure. The invariant speed c (set to 1 in our units) emerges as the ratio of phase stiffness to temporal correlation decay rate.

10.3 Modification by Defects

Near a massive topological defect, the amplitude A is suppressed and the phase stiffness $\kappa = \alpha A^2$ is reduced. This modifies the null cone structure of the emergent

metric: light cones tilt inward toward the defect, consistent with the bending of light by a gravitating body. In the extreme case where $A \rightarrow 0$ (the interior of a “black hole” defect, Section 26), the null cone tips over completely, and the causal structure develops a horizon. The analogy to the Kerr metric’s ergoregion is apt: within the ergoregion, the time-translation Killing vector becomes spacelike; in LPST terms, the phase-update direction rotates relative to the label-space coordinates.

11. Time as Phase-Update Ordering

In LPST, time is not a fundamental parameter but an emergent concept defined by the internal dynamics of the substrate.

11.1 Proper Time from Phase Activity

We define the emergent proper time interval along a worldline (a curve in label space) as:

$$d\tau \propto \sqrt{\kappa(x)} \cdot |\partial\phi| \quad (26)$$

where $\kappa(x) = \alpha |L(x)|^2$ is the local phase stiffness. This definition captures the physical intuition that time “passes” at a rate determined by the local dynamical activity of the substrate. Regions of high phase stiffness and rapid phase evolution experience more proper time per unit label-space parameter; regions of low phase stiffness experience less.

11.2 Gravitational Time Dilation

Near a massive topological defect, the amplitude A is suppressed relative to the vacuum value A_0 , so $\kappa(x) = \alpha A(x)^2 < \alpha A_0^2 = \kappa_0$. The ratio of proper time rates at two points x_1 (near the defect) and x_2 (far from the defect) is:

$$d\tau_1/d\tau_2 = \sqrt{\kappa(x_1)/\kappa(x_2)} = A(x_1)/A(x_2) \quad (27)$$

In the weak-field limit (Section 20), the amplitude profile around a static spherically symmetric defect of mass M takes the form $A(r) \approx A_0\sqrt{1 - 2GM/r}$, so:

$$d\tau_1/d\tau_2 \approx \sqrt{1 - 2GM/r} \quad (28)$$

which is precisely the Schwarzschild time dilation factor. Gravitational time dilation emerges naturally from the modulation of phase stiffness by matter.

11.3 Relation to the Problem of Time

The “problem of time” in quantum gravity [25] arises because the Wheeler-DeWitt equation $H\Psi = 0$ is timeless: the total Hamiltonian constraint of canonical gravity annihilates the physical state, leaving no external time parameter for dynamics. In LPST, this problem is dissolved: time is not an external parameter but an internal, relational quantity defined by the substrate’s own phase dynamics. The analogy is to the Page-Wootters mechanism [31], in which a subsystem of the universe serves as a clock for the rest. In LPST, the phase field itself is the universal clock, and different regions of the substrate define their own local proper times through their local phase activity. The global timelessness of the fundamental description (there is no external time in the substrate) is consistent with the local emergence of time in the effective spacetime.

12. Lorentz Invariance as Emergent Symmetry

12.1 Derivation from Correlation Symmetry

In the vacuum state, the correlation kernel $C(x,y)$ depends only on the separation $\Delta x = x - y$ and takes the Lorentz-invariant form $C(\Delta x) = A_0^2 \exp(-\eta_{\mu\nu}\Delta x^\mu\Delta x^\nu/(2\xi^2))$, where $\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$ is the Minkowski metric. The emergent metric (24) therefore inherits the symmetries of this kernel: the Lorentz group $SO(3,1)$, consisting of rotations and boosts, preserves the form of C and hence of $g_{\mu\nu}$. Lorentz invariance is thus an emergent symmetry of the vacuum correlation structure.

12.2 Emergent Speed of Light

The emergent speed of light c is determined by the ratio of the temporal and spatial correlation scales in the vacuum kernel. If C depends on $t^2/\xi_t^2 - |\mathbf{x}|^2/\xi_s^2$, then $c =$

ξ_s/ξ_t . In the vacuum, this ratio is fixed by the parameters α and A_0 , giving a universal, isotropic speed of light. We set $c = 1$ by choice of units.

12.3 Lorentz Violation at Ultra-High Energies

Lorentz invariance is exact in the low-energy, long-wavelength regime where the Gaussian approximation to $C(x,y)$ holds. At energies approaching the substrate scale $E \sim 1/\xi$ (which we tentatively identify with the Planck energy $E_P \sim 10^{19}$ GeV), higher-order terms in the Taylor expansion of $\ln|C|$ become significant, and the emergent metric develops energy-dependent corrections. This predicts Lorentz invariance violation (LIV) of the form:

$$v(E) = 1 - \eta(E/E_P)^n + O((E/E_P)^{n+1}) \quad (29)$$

where $n = 1$ or 2 depending on the symmetry properties of the substrate, and η is an order-one coefficient. Current constraints from Fermi-LAT observations of gamma-ray bursts [32] limit $\eta < O(1)$ for $n = 1$ (linear LIV) at the Planck scale, while $n = 2$ (quadratic LIV) remains largely unconstrained. This is a testable prediction of LPST: next-generation gamma-ray observatories such as the Cherenkov Telescope Array (CTA) will probe the quadratic regime.

13. Matter as Topological Defects — General Theory

In LPST, matter particles are not fundamental fields but stable topological solitons—localized, finite-energy configurations of the substrate that cannot be continuously deformed to the vacuum. Their stability is guaranteed by topology: the field configuration wraps around non-contractible cycles of the target space, and this wrapping is characterized by integer-valued topological charges that are exactly conserved.

13.1 Topological Currents and Charges

For the substrate field taking values in $\mathbb{C} \times S^2 \cong (S^1 \times \mathbb{R}_+) \times S^2$, the topological current can be defined as:

$$j^\mu = (1/(8\pi^2)) \epsilon^{\mu\nu\rho\sigma} \epsilon_{abc} u^a \partial_\nu u^b \partial_\rho u^c \partial_\sigma \varphi \quad (30)$$

This current is identically conserved, $\partial_\mu j^\mu = 0$, as a consequence of the antisymmetry of the epsilon tensors. The conserved topological charge is:

$$Q = \int j^0 d^3x \in \mathbb{Z} \quad (31)$$

The integer quantization of Q follows from the quantization of the winding numbers of the maps $\varphi: S^1 \rightarrow S^1$ and $\mathbf{u}: S^2 \rightarrow S^2$. This topological charge is the LPST origin of baryon number, lepton number, and other discrete quantum numbers of the Standard Model.

13.2 Evasion of Derrick's Theorem

Derrick's theorem [26] states that static, finite-energy solutions of scalar field theories with standard kinetic terms do not exist in spatial dimensions $d \geq 2$. The proof proceeds by showing that under a rescaling $x \rightarrow \lambda x$, the kinetic energy scales as λ^{2-d} while the potential energy scales as λ^{-d} , so no stationary point of the total energy exists for $d \geq 2$. LPST evades Derrick's theorem via the topological stabilizer term $K(\varphi, \mathbf{u})$: the Skyrme term contributes an energy that scales as λ^{-2} (more negative power of λ than the standard kinetic term), providing a balancing force that stabilizes the soliton at a finite size $R \sim \sqrt{(\gamma/\alpha)}$. This is the same mechanism by which Skyrme's original model stabilizes baryons [27].

13.3 Mass as Trapped Phase Energy

The mass of a topological defect is defined as its total energy in the rest frame:

$$M = \int d^3x [\alpha A_0^2 (\nabla\varphi)^2 + \beta A_0^2 (\nabla\mathbf{u})^2 + \gamma K + \lambda(A^2 - A_0^2)^2] \quad (32)$$

This integral is dominated by the phase gradient and polarization gradient contributions. Mass is thus literally *trapped phase energy*—the energy stored in the non-trivial winding of the substrate fields. This is the LPST realization of $E = mc^2$: mass is not a fundamental property but a measure of the total phase distortion energy locked into a topological configuration. The conservation of mass (in non-relativistic contexts) follows from the conservation of topological charge: since the

winding cannot unwind without encountering another defect of opposite charge, the energy content is trapped indefinitely.

13.4 Defect Interactions

Two defects interact via the overlap of their phase and polarization profiles. At large separations $r \gg \xi$, the interaction energy falls off as a power law or exponentially, depending on the topological sector. Phase vortices of the same sign repel (their phase gradients constructively interfere, increasing the total energy), while vortex-antivortex pairs attract (their gradients partially cancel). This reproduces the qualitative features of electromagnetic interactions between charged particles. The detailed force law is derived in Sections 17 and 19.

14. Classification of Topological Defects

The richness of the target space $S^1 \times S^2$ of the substrate fields $(\varphi, \mathbf{u})$ supports a variety of topological defects, classified by the homotopy groups of the target space.

Defect Type	Topological Invariant	Homotopy Group	Emergent Property	Physical Correspondence
Phase vortex	$\oint \nabla\varphi \cdot d\mathbf{l} = 2\pi n$	$\pi_1(S^1) = \mathbb{Z}$	Quantized charge	Charged leptons
Skyrmion	$\deg(\mathbf{u}: S^2 \rightarrow S^2)$	$\pi_2(S^2) = \mathbb{Z}$	Half-integer spin	Fermions
Hopfion (knotted soliton)	Hopf invariant $H[\mathbf{u}]$	$\pi_3(S^2) = \mathbb{Z}$	Mass / baryon number	Baryons
Braided defects	Braid group element	B_n	Non-Abelian statistics	Anyonic excitations
Linked vortex rings	Linking number	—	Bound state stability	Mesons
Monopole-like (hedgehog)	Hedgehog charge	$\pi_2(S^2) = \mathbb{Z}$	Magnetic charge	Magnetic monopoles

14.1 Phase Vortices and Electric Charge

A phase vortex is a line-like defect around which the phase φ winds by $2\pi n$, where $n \in \mathbb{Z}$ is the winding number. The core of the vortex (where $A \rightarrow 0$) is a one-dimensional line in three-dimensional space; in spacetime, it traces out a worldsheet. The winding number n is identified with the electric charge (in appropriate units). Vortex-antivortex pairs ($\pm n$) can annihilate, releasing their phase gradient energy as radiation—the substrate analogue of particle-antiparticle annihilation.

14.2 Skyrmions and Fermions

A skyrmion is a localized configuration of the polarization field $\mathbf{u}$ in which the map $\mathbf{u}: \mathbb{R}^3 \rightarrow S^2$ (with the boundary condition $\mathbf{u} \rightarrow \mathbf{u}_0$ at spatial infinity) wraps the target sphere a non-trivial number of times. The degree of this map is the skyrmion number $B \in \mathbb{Z}$, which is identified with the baryon number in the nuclear physics context [27]. In LPST, skyrmions acquire fermionic statistics because a 2π rotation of the skyrmion in physical space induces a sign change in the substrate field, yielding a factor of $(-1)^{2B} = -1$ for $B = \text{half-integer}$. This is the topological origin of the spin-statistics connection (Section 15).

14.3 Hopfions and Baryonic Matter

The Hopf invariant classifies maps from $S^3 \rightarrow S^2$ and takes integer values. In the substrate, a hopfion is a knotted soliton in which the preimages of distinct points on S^2 are linked circles in $\mathbb{R}^3$. This linking prevents the configuration from unwinding and provides additional topological stability. The hopfion number is identified with the baryon number in LPST. The mass spectrum of hopfions depends on the parameters α, β, γ and the specific form of K , and reproducing the observed baryon mass spectrum is a key challenge for the theory (Section 31).

14.4 The Analogy to Skyrme's Model

Skyrme proposed in 1962 [27] that baryons are topological solitons of the pion field, with the baryon number identified as the skyrmion number. This proposal was vindicated in the large- N_c limit of QCD, where baryons do emerge as solitons of the chiral Lagrangian [33]. LPST extends this idea to the most fundamental level: *all*

matter, not just baryons, consists of topological solitons—but of the substrate field rather than the pion field. The pion-field skyrmions of nuclear physics are then understood as effective, coarse-grained descendants of the fundamental substrate defects.

15. Spin, Statistics, and the Braid Group

15.1 The Spin-Statistics Connection

One of the deepest results in quantum field theory is the spin-statistics theorem: particles with integer spin obey Bose-Einstein statistics, and particles with half-integer spin obey Fermi-Dirac statistics [34]. In LPST, this connection arises from the topology of the configuration space of defects.

Consider a skyrmion of unit topological charge $B = 1$. A physical rotation of the skyrmion by angle 2π about any axis is equivalent to a gauge transformation in the substrate: it maps the configuration $\mathbf{u}(\mathbf{x})$ to a configuration $\mathbf{u}'(\mathbf{x})$ that is related to the original by a topologically non-trivial loop in the configuration space. This loop contributes a phase factor $\exp(i\pi B)$ to the quantum amplitude. For $B = 1$ (or any odd integer), this factor is -1 , implying fermionic statistics. For $B = 0$ or any even integer, the factor is $+1$, implying bosonic statistics.

15.2 Anyons in 2+1 Dimensions

In systems confined to two spatial dimensions (relevant to certain condensed matter systems and potentially to boundary theories of LPST), the relevant group governing the exchange of identical defects is the braid group B_n rather than the symmetric group S_n . This allows for anyonic statistics: the exchange of two identical defects yields a phase factor $\exp(i\theta)$ with arbitrary θ , not just 0 or π . LPST naturally accommodates braided defect configurations (Table, Section 14), which may be relevant to topological quantum computation and may have observable signatures in analogue systems.

15.3 Extension to 3+1 Dimensions

In three spatial dimensions, the braid group on n particles reduces to the symmetric group S_n , which admits only two one-dimensional representations: the trivial (bosonic) and the sign (fermionic) representation. This recovers the standard boson/fermion dichotomy of three-dimensional physics. The restriction from continuous θ (anyons) to discrete $\theta = 0$ or π (bosons or fermions) is thus a consequence of the dimensionality of space—which, in LPST, is itself emergent from the correlation structure. This raises the intriguing possibility that the three-dimensionality of space is related to the requirement that only bosons and fermions exist, providing a novel perspective on the dimension of space.

16. Gauge Symmetries from Internal Polarization

16.1 The Mechanism

Gauge symmetries in LPST emerge from the isometries of the internal polarization manifold M_{int} . When $\mathbf{u}$ takes values in S^2 , the isometry group of the target space is $SO(3) \cong SU(2)/\mathbb{Z}_2$. Local rotations of $\mathbf{u}$ that leave the Lagrangian invariant generate an emergent gauge symmetry.

To see this explicitly, consider a local rotation $\mathbf{u}(\mathbf{x}) \rightarrow R(\mathbf{x}) \mathbf{u}(\mathbf{x})$, where $R(\mathbf{x}) \in SO(3)$ varies from point to point. The kinetic term $(\partial_\mu \mathbf{u})^2$ is not invariant under this transformation because the derivative hits $R(\mathbf{x})$. However, we can define a covariant derivative $D_\mu \mathbf{u} = \partial_\mu \mathbf{u} + A_\mu \times \mathbf{u}$, where A_μ is a connection (gauge field) valued in the Lie algebra $\mathfrak{so}(3)$. The combination $(D_\mu \mathbf{u})^2$ is gauge-invariant by construction. In LPST, the gauge field A_μ is not an additional fundamental field; it is induced by the topology of the defect configurations and can be expressed in terms of derivatives of $\mathbf{u}$:

$$A_\mu^a = \epsilon^{abc} u^b \partial_\mu u^c / (1 + \mathbf{u} \cdot \mathbf{u}_0) \quad (33)$$

This is the canonical connection on S^2 pulled back to spacetime by the map $\mathbf{u}(\mathbf{x})$. Its curvature (field strength) is the emergent gauge field strength.

16.2 Gauge Coupling Constant

The gauge coupling constant g is determined by the substrate parameters. From the polarization kinetic term $\beta A_0^2 (D_\mu \mathbf{u})^2$, we identify $g \sim 1/\sqrt{\beta A_0^2}$. The fine-structure constant $\alpha_{\text{em}} = g^2/(4\pi)$ is thus a derived quantity in LPST, determined by the ratio of the polarization coupling β to the vacuum amplitude A_0 . Computing its value from first principles requires solving the self-consistent equations for the vacuum state, which is a major open challenge.

16.3 Analogy to Kaluza-Klein Theory

This mechanism for generating gauge fields is structurally analogous to the Kaluza-Klein construction [35], in which gauge fields arise from the metric of extra compact spatial dimensions. In LPST, the “extra dimensions” are not spatial directions but internal field-space dimensions (the target manifold M_{int}). This avoids the need for physical extra dimensions and their associated moduli stabilization problems, while retaining the elegant unification of gauge and gravitational physics that Kaluza-Klein theory achieves.

17. Electromagnetism as Polarization Waves

17.1 Linearized Dynamics and Maxwell’s Equations

We linearize the polarization equation (17) around the vacuum $\mathbf{u} = \mathbf{u}_0 + \delta\mathbf{u}$, with $\delta\mathbf{u}$ perpendicular to $\mathbf{u}_0$ (to maintain the constraint $|\mathbf{u}| = 1$). To leading order in $\delta\mathbf{u}$, and setting $A = A_0 = \text{const}$, the equation of motion reduces to:

$$\nabla^2(\delta\mathbf{u}) = 0 \quad (34)$$

This is a massless wave equation for two transverse polarization modes—exactly the structure of free electromagnetic radiation. The two independent components of $\delta\mathbf{u}$ (perpendicular to $\mathbf{u}_0$) correspond to the two polarization states of the photon.

17.2 Identification of the Field Strength

The electromagnetic field strength tensor is identified with the curvature of the emergent U(1) connection arising from the phase sector:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad (35)$$

where A_μ is the U(1) gauge potential derived from the phase gradient: $A_\mu \propto \partial_\mu \varphi$. The Maxwell equations $\partial_\mu F^{\mu\nu} = J^\nu$ follow from the phase field equation (15), with the source current J^ν provided by the topological vortex current.

17.3 Charge Conservation and the Coulomb Potential

Electric charge conservation is a direct consequence of the topological conservation of vortex winding number. The winding number $n = (1/2\pi) \oint \nabla \varphi \cdot d\mathbf{l}$ is an integer and cannot change under continuous evolution of the fields. Charge is thus exactly conserved in LPST, not by fiat or by Noether's theorem, but by topological necessity.

The interaction energy between two vortices of winding numbers n_1 and n_2 separated by a large distance r is computed from the overlap of their phase gradients. In three spatial dimensions, the phase gradient of a single vortex falls off as $1/r$, and the interaction energy is:

$$E_{\text{int}} \propto n_1 n_2 / r \quad (36)$$

This is the Coulomb potential, with the charge proportional to the winding number. The long-range $1/r$ form is a consequence of the masslessness of the phase mode (Goldstone boson of the broken U(1) symmetry).

18. Non-Abelian Gauge Fields and the Strong Interaction

18.1 Extension to SU(3)

To accommodate the strong interaction, we generalize the internal manifold M_{int} from S^2 to a space with a larger isometry group. If M_{int} has an SU(3) isometry—for example, if $M_{\text{int}} = \text{SU}(3)/\text{U}(1)^2$ or a suitable quotient space—then the construction of

Section 16 generates SU(3) gauge fields from the gradients of $\mathbf{u}$. The color charge of quarks corresponds to the orientation of the polarization field in the internal SU(3) directions.

18.2 Confinement from Topology

Confinement—the empirical fact that quarks are never observed in isolation—arises naturally in LPST. Braided and linked defect configurations have a linear energy cost for separation: pulling two linked vortex rings apart stretches the linking region, creating a tube of distorted polarization field whose energy grows linearly with the separation distance d :

$$E_{\text{tube}} \propto \sigma \cdot d \quad (37)$$

where σ is the string tension, determined by β , A_0 , and the geometry of the flux tube cross-section. This is the LPST analogue of the QCD flux tube [36], and the linear potential reproduces the empirically observed Regge trajectories of hadronic resonances. When the energy stored in the tube exceeds the mass of a defect-antidefect pair, the tube breaks by creating a new pair—the analogue of string breaking and hadronization in QCD.

18.3 The Standard Model Gauge Group

The precise identification of M_{int} such that its isometry group yields exactly the Standard Model gauge group $G_{\text{SM}} = \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ is a major open problem in LPST. Several candidate manifolds exist—for example, $M_{\text{int}} = \text{CP}^2 \times \text{S}^2$ has isometry group $\text{SU}(3) \times \text{SO}(3)$, which contains the required structure. However, the detailed identification, including the correct hypercharge assignments and the mechanism for electroweak symmetry breaking, requires further development. We regard this as the most important direction for future work in the gauge sector of LPST.

19. Gravity as Self-Induced Refraction

This section develops the gravitational sector of LPST, showing that all gravitational phenomena arise from the modulation of phase stiffness by matter.

19.1 Phase Stiffness and the Refractive Index

Localized topological defects modify the local amplitude of the substrate: $A(x) \neq A_0$ in the vicinity of a defect. Since the phase stiffness is:

$$\kappa(x) = \alpha |L(x)|^2 = \alpha A(x)^2 \quad (38)$$

a defect that suppresses A creates a region of reduced phase stiffness. Phase perturbations (light) propagating through this region travel with an effective speed that depends on $\kappa(x)$, defining an effective refractive index:

$$n(x) = \sqrt{\kappa_0/\kappa(x)} = A_0/A(x) \quad (39)$$

where $\kappa_0 = \alpha A_0^2$ is the vacuum phase stiffness. In the weak-field regime where $A(x) \approx A_0(1 - \Phi(x)/c^2)$ with Φ the Newtonian potential, the refractive index becomes:

$$n(x) \approx 1/(1 - \Phi/c^2) \approx 1 + \Phi(x)/c^2 + \dots \quad (40)$$

19.2 The Effective Metric

Perturbations of the phase field propagate on an effective metric determined by the local phase stiffness. This effective metric takes the Gordon form [21]:

$$g^{\mu\nu}_{\text{eff}} \sim \kappa(x)^{-1} \eta^{\mu\nu} + \Delta^{\mu\nu} \quad (41)$$

where $\eta^{\mu\nu}$ is the background Minkowski metric and $\Delta^{\mu\nu}$ encodes corrections from the specific defect profile. This effective metric produces all standard gravitational effects:

- **Geodesic bending:** Paths of extremal correlation (null geodesics of g_{eff}) curve around massive defects, reproducing gravitational lensing.
- **Gravitational redshift:** Phase waves propagating from low- κ to high- κ regions lose energy (redshift), because the phase oscillation rate changes with κ .
- **Time dilation:** The rate of proper time (26) depends on κ , producing gravitational time dilation as in Eq. (28).

- **Shapiro delay:** Phase waves passing near a massive defect travel through a region of higher refractive index and are delayed relative to waves traveling through vacuum.

19.3 Physical Picture

The physical picture is vivid: a massive defect is a concentration of phase energy that locally thins the substrate (suppresses A). This thinned region acts like a lens, refracting passing phase waves. The refraction is attractive because the refractive index increases toward the defect. All “gravitational” effects are refractive effects in the phase substrate. There is no fundamental gravitational field, no graviton, and no spacetime curvature at the fundamental level. Curvature is an effective description of the refraction pattern, valid in the regime where a smooth effective metric can be defined.

20. Linearized Gravity and the Newtonian Limit

20.1 Static Spherically Symmetric Defect

Consider a single static, spherically symmetric topological defect of total phase energy M located at the origin. The amplitude profile, determined by the amplitude equation (14) in the weak-field regime, takes the form:

$$A(r) \approx A_0(1 - GM/(r c^2)) \quad (42)$$

where G is to be determined in terms of substrate parameters. The phase stiffness is then $\kappa(r) = \alpha A_0^2 (1 - 2GM/(rc^2) + \dots)$. From Eq. (41), the effective metric in isotropic coordinates is:

$$ds^2 \approx -(1 - 2GM/r) dt^2 + (1 + 2GM/r)(dx^2 + dy^2 + dz^2) \quad (43)$$

This is the linearized Schwarzschild metric in isotropic coordinates, valid for $r \gg 2GM$.

20.2 Geodesic Motion and Newton's Law

A test defect (a defect of negligible mass $m \ll M$) moving in the effective metric (43) follows geodesics. In the non-relativistic limit ($v \ll c$), the geodesic equation reduces to:

$$d^2\mathbf{x}/dt^2 = -\nabla\Phi(r), \quad \Phi(r) = -GM/r \quad (44)$$

which is Newton's law of gravitation: $\mathbf{F} = -GMm \hat{\mathbf{r}}/r^2$. The force is attractive, central, and inversely proportional to the square of the distance, with the coupling constant G determined by:

$$G \propto 1/(\alpha A_0^2) \quad (45)$$

This identification has a clear physical meaning: a stiffer substrate (α large) or a denser substrate (A_0 large) produces weaker gravity, because the fractional perturbation of the phase stiffness by a given mass is smaller. The equivalence principle is automatically satisfied because the effective metric couples universally to all forms of energy—every defect, regardless of its internal structure, propagates on the same $g_{\mu\text{veff}}$.

21. Recovery of General Relativity

21.1 Self-Consistency and the Einstein Equations

The full recovery of general relativity in LPST proceeds from the self-consistency requirement: the metric appearing in the substrate action (5) must equal the metric derived from the correlations of the resulting field configurations via Eq. (24). This self-consistency condition, combined with the diffeomorphism invariance of the correlation-to-metric construction, constrains the dynamics of the emergent metric.

The argument proceeds in analogy with Jacobson's thermodynamic derivation [5]. Consider a local Rindler horizon at an arbitrary spacetime point. The substrate field's phase stiffness defines a local temperature (the Unruh temperature $T = a/(2\pi)$, where a is the proper acceleration). The entropy is proportional to the area of the correlation-defined horizon (the number of independent phase degrees of freedom on the horizon surface). The energy flux across the horizon is the substrate

energy-momentum tensor integrated over the horizon generators. Demanding that the Clausius relation $\delta Q = T dS$ holds for all local Rindler horizons yields the Einstein equation:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu} \quad (46)$$

21.2 The Cosmological Constant

The cosmological constant Λ arises naturally as the vacuum energy of the substrate:

$$\Lambda = 8\pi G \cdot V(A_0) = 8\pi G \cdot \lambda \cdot 0 = 0 \quad (47)$$

At tree level, the Mexican-hat potential vanishes at its minimum, giving $\Lambda = 0$. Quantum corrections (fluctuations of A and ϕ around the vacuum) generate a non-zero vacuum energy. The observed smallness of the cosmological constant may then be related to the near-cancellation between different topological sectors of the substrate—a possibility explored in Section 27.

21.3 Beyond General Relativity

LPST predicts corrections to GR at high curvature scales (curvature radius approaching the correlation length ξ). These corrections take the form of higher-derivative terms in the effective gravitational action:

$$S_{\text{grav}} = \int d^4x \sqrt{(-g)} [R/(16\pi G) + c_1 R^2 + c_2 R_{\mu\nu} R^{\mu\nu} + \dots] \quad (48)$$

where the coefficients c_1, c_2 are determined by the substrate parameters and are of order ξ^2 . These corrections are negligible at laboratory scales but may be significant near black hole singularities and in the very early universe.

22. Recovery of Quantum Mechanics

22.1 The Path Integral from Phase Consistency

The phase consistency principle (Postulate 3) requires summing over all consistent phase configurations connecting specified initial and final substrate states. This is

precisely the Feynman path integral [37]. The weight of each configuration is determined by its phase:

$$\langle \text{final} | \text{initial} \rangle = \int [\text{DL}] \exp(i S[\text{L}]/\hbar) \quad (49)$$

where the integral is over all phase-consistent field configurations interpolating between the initial and final states. The action $S[\text{L}]$ is the substrate action (5), and $\hbar$ is determined by the fundamental scale of phase quantization: the minimum quantum of action is set by the quantization of the topological charges, giving $\hbar \sim \alpha A_0^2 \xi^2$.

22.2 The Born Rule

The Born rule—that the probability of an outcome is $|\langle \text{final} | \text{initial} \rangle|^2$ —follows from a phase-space volume argument. Among all phase-consistent configurations contributing to the path integral, the number of configurations leading to outcome a with amplitude ψ_a is proportional to $|\psi_a|^2$. This is because destructive interference suppresses the contributions of configurations with random relative phases, and constructive interference enhances those that are phase-aligned. The resulting probability is $P(a) = |\psi_a|^2 / \sum_b |\psi_b|^2$.

22.3 The Schrödinger Equation

In the low-energy, non-relativistic limit, the path integral (49) reduces to the quantum mechanics of point particles (topological defects with size $\ll \xi$). The standard derivation of the Schrödinger equation from the path integral [37] applies: for a defect of mass M in a slowly varying potential $V(x)$, the path integral yields the propagator satisfying:

$$i\hbar \partial\psi/\partial t = -(\hbar^2/(2M)) \nabla^2\psi + V(x)\psi \quad (50)$$

where $\psi(x,t)$ is the effective wavefunction of the defect's center-of-mass degree of freedom.

22.4 Measurement and Decoherence

In LPST, a “measurement” occurs when a topological defect (the observer/apparatus) becomes phase-correlated with another defect (the system).

The correlation established by the measurement interaction causes the off-diagonal elements of the reduced density matrix of the system to decay—decoherence. This occurs because the observer defect has a vast number of internal degrees of freedom (it is a macroscopic aggregate of many topological defects), and tracing over these unmeasured degrees of freedom destroys the phase coherence between different branches of the superposition. The decoherence time scale is determined by the rate at which phase correlations between the system and environment develop, which depends on the coupling strength and the size of the environment. There is no need for a separate “collapse postulate”: the appearance of definite outcomes is a consequence of decoherence within the unitary evolution of the substrate.

23. Mass-Energy Equivalence and Field Quantization

23.1 Derivation of $E = mc^2$

The rest energy of a topological defect is its total phase gradient energy in the rest frame, Eq. (32). We denote this $M c^2$ (restoring c for clarity). For a defect boosted to velocity v , the Lorentz transformation of the phase gradient $\partial_\mu \varphi$ yields:

$$\partial_0 \varphi \rightarrow \gamma(\partial_0 \varphi - v \partial_1 \varphi), \quad \partial_1 \varphi \rightarrow \gamma(\partial_1 \varphi - v \partial_0 \varphi) \quad (51)$$

where $\gamma = 1/\sqrt{1 - v^2/c^2}$ is the Lorentz factor. The total energy (integral of T_{00}) of the boosted defect is:

$$E = \gamma M c^2 \quad (52)$$

confirming the relativistic energy-momentum relation. At $v = 0$, $E = M c^2$: the rest energy equals the mass times c^2 . This is not postulated but derived from the Lorentz transformation properties of the phase gradient.

23.2 Quantization of Excitations

The discreteness of topological charges ensures that excitations of the substrate are inherently quantized. Phase vortices carry integer winding number n , skyrmions carry integer degree B , and hopfions carry integer Hopf invariant H . These quantum

numbers cannot change continuously—they jump in integer steps. This provides a topological foundation for the quantization of charge, spin, and baryon number. The phase sector supports integer-quantized excitations (bosonic), while the polarization sector supports half-integer-quantized excitations (fermionic), as shown in Section 15.

23.3 Planck's Constant from the Substrate

The Planck constant $\hbar$ emerges as the minimum quantum of phase action. The phase φ is a periodic variable: φ and $\varphi + 2\pi$ are physically identical. The action associated with a single phase winding around a minimal loop is:

$$S_{\min} = \alpha A_0^2 \oint (\nabla\varphi)^2 dV \sim \alpha A_0^2 (2\pi)^2 \xi \quad (53)$$

This minimal action quantum is identified with $2\pi\hbar$, giving $\hbar \sim \alpha A_0^2 \xi \cdot 2\pi$. The Planck constant is thus a derived quantity, set by the phase stiffness, vacuum amplitude, and correlation length of the substrate.

24. Thermodynamics and Entropy from Phase Coherence

24.1 Entropy as Loss of Phase Coherence

Thermodynamic entropy in LPST is identified with the loss of accessible phase information. When an observer (a collection of topological defects) interacts with only a subsystem of the substrate, the unobserved degrees of freedom are traced over, yielding a reduced density matrix ρ for the observed subsystem. The von Neumann entropy:

$$S = -k_B \text{Tr}(\rho \ln \rho) \quad (54)$$

quantifies the observer's ignorance of the phase correlations between the observed subsystem and the unobserved environment. This is a relational, observer-dependent quantity, consistent with the information-theoretic interpretation of thermodynamic entropy.

24.2 Temperature and the Unruh Effect

Temperature emerges as the rate of phase decoherence experienced by an accelerating defect. An accelerating defect perceives the vacuum phase fluctuations as a thermal bath because its correlation horizon (the Rindler horizon) excludes part of the phase configuration. The temperature is:

$$T = \hbar a / (2\pi c k_B) \quad (55)$$

where a is the proper acceleration. This is the Unruh temperature [23], derived here from the substrate's correlation structure rather than from quantum field theory on a fixed background.

24.3 Bekenstein-Hawking Entropy

For a black hole defect (a region where $A \rightarrow 0$ and a correlation-defined horizon forms, Section 26), the entropy is determined by the number of independent phase degrees of freedom on the horizon surface. The correlation length ξ sets the fundamental scale: the number of independent phase cells on a horizon of area A is $N \sim A/\xi^2$, so:

$$S_{BH} = k_B N \sim k_B A/\xi^2 \quad (56)$$

Identifying ξ^2 with $4G\hbar/c^3$ (the Planck area, up to numerical factors), we recover $S_{BH} = k_B c^3 A/(4G\hbar)$, the Bekenstein-Hawking formula [1,2].

24.4 The Second Law

The second law of thermodynamics follows from the irreversibility of phase decoherence in the thermodynamic limit. While the substrate's microscopic dynamics are unitary (phase information is globally conserved), the coarse-grained description accessible to any finite collection of defects (observer) inevitably loses track of phase correlations as the number of interacting degrees of freedom grows. This is the standard mechanism of decoherence-induced entropy increase, applied to the substrate.

25. Cosmological Implications

25.1 The Big Bang as a Phase Transition

In LPST, the Big Bang is not a singularity in spacetime but a phase transition of the substrate from a maximally coherent, maximally symmetric configuration to the current decoherent state. Before the transition, the substrate was in a state of perfect phase alignment: $C(x,y) = A_0^2$ for all x, y , and the emergent metric was undefined (or trivially zero-dimensional, since all points were maximally correlated and hence at zero effective distance). The phase transition broke this perfect coherence and initiated the growth of spatial structure.

25.2 Cosmic Expansion

Cosmic expansion is the growth of the emergent spatial geometry as phase correlations evolve. As the substrate transitions from the initial maximally coherent state, the correlation kernel $C(x,y)$ develops structure: it begins to decay with separation, defining an increasing correlation length $\xi(t)$ and a growing effective spatial volume. The scale factor $a(t)$ of the Friedmann-Lemaître-Robertson-Walker metric is identified with the ratio of the correlation length at time t to its value at some reference time. The Friedmann equations follow from the self-consistent coupling of the metric to the substrate energy-momentum tensor, as in Section 21.

25.3 Inflation as Phase Ordering

Inflation—a period of exponential expansion in the very early universe—can be modeled in LPST as a period of rapid phase ordering, analogous to domain formation in a ferromagnet after rapid cooling through its Curie temperature. During the phase transition, the amplitude A rapidly settles to its vacuum value A_0 (this is the analogue of the inflaton field rolling to the minimum of its potential), releasing the potential energy $V(|L|) - V(A_0)$ as an effective cosmological constant that drives exponential expansion. The ordering of the phase and polarization fields proceeds via the growth and coalescence of coherence domains, and inflation ends when the ordering is complete.

25.4 The Horizon Problem

The horizon problem of standard cosmology—the question of why the cosmic microwave background is so uniform despite distant regions being causally disconnected in the standard picture—is dissolved in LPST. The initial state of the substrate is a single, maximally coherent field configuration. All regions of the substrate were phase-correlated from the outset, because the initial state had $C(x,y) = A_0^2$ everywhere. The uniformity of the CMB is a remnant of this initial coherence.

25.5 Primordial Perturbations

Primordial density perturbations arise from quantum fluctuations of the phase and amplitude fields during the phase-ordering epoch. These fluctuations are stretched to macroscopic scales by the inflationary expansion and subsequently seed the large-scale structure of the universe. The power spectrum is determined by the two-point function of the phase fluctuations, which, for a slowly ordering system, yields a nearly scale-invariant spectrum:

$$P(k) \propto k^{n_s-1}, \quad n_s \approx 1 - 2/N \quad (57)$$

where N is the number of e-folds of inflation. For $N \sim 60$, this gives $n_s \approx 0.97$, consistent with Planck satellite measurements [38].

26. Black Holes, Horizons, and Information

26.1 Black Holes as Extreme Phase-Stiffness Defects

A black hole in LPST is a region where the phase stiffness $\kappa(x) = \alpha A(x)^2$ is driven to zero (or becomes negative). This occurs when the density of trapped phase energy is so high that the amplitude A is suppressed to zero, creating a “hole” in the substrate. The boundary of this region—where $A(r) = 0$ and κ drops to zero—is the correlation-defined horizon.

26.2 The Horizon as a Correlation Boundary

The event horizon is the surface where the correlation kernel $C(x,y)$ drops to zero for points x inside and y outside the horizon. This means that the emergent distance between interior and exterior points is infinite ($\ln(0) = -\infty$ in Eq. (19)), and no phase information can propagate from inside to outside. Inside the horizon, the phase gradients become spacelike: the “radial” direction becomes temporal, and the defect is inevitably drawn toward the center—the substrate analogue of the singularity.

26.3 The Information Paradox

The black hole information paradox [39] asks: if information falls into a black hole and the black hole subsequently evaporates via Hawking radiation, is the information lost? In LPST, information is encoded in the topological structure of the substrate, which is globally defined. Topological charges (winding numbers, skyrmion numbers, Hopf invariants) are globally conserved and cannot be destroyed by local dynamics. The apparent loss of information at the horizon is an artifact of the emergent geometric description: the effective metric has a singularity, but the underlying substrate field is smooth (or at least topologically well-defined) everywhere. The full substrate description preserves unitarity because the global phase configuration evolves unitarily under the phase consistency principle.

In practice, the information escapes during the evaporation process via subtle correlations in the Hawking radiation—correlations that are invisible in the effective semiclassical description but are present in the full substrate theory. This is consistent with the Page curve [40] and with recent results from the gravitational path integral [41].

26.4 Observational Signatures

LPST predicts modifications to the Hawking radiation spectrum near the Planck scale, where the substrate structure becomes important. Additionally, the correlation-defined horizon may exhibit “echoes” in gravitational wave ringdown signals: reflections of gravitational waves from the inner structure of the horizon, which in LPST has a finite (though extremely small) width set by the correlation

length ξ . These echoes are potentially detectable by current and future gravitational wave observatories [42].

27. Dark Matter and Dark Energy

27.1 Dark Matter from Diffuse Defect Configurations

LPST offers a natural candidate for dark matter: diffuse topological defect configurations that interact gravitationally but have suppressed electromagnetic coupling. Specifically, consider a condensate of skyrmions or a “phase fog”—a spatially extended, low-density configuration of the polarization field with non-trivial topology. Such configurations modify the phase stiffness $\kappa(x)$ (and hence the emergent metric), producing gravitational effects, but they do not carry vortex winding number (quantized electric charge) and therefore do not couple to electromagnetism.

The mass distribution of dark matter defects may exhibit characteristic spatial correlations that differ from those of Weakly Interacting Massive Particles (WIMPs) or other standard dark matter candidates. In particular, the topological stability of the defect configurations may impose minimum-size constraints on dark matter halos, potentially addressing the “core-cusp” and “too-big-to-fail” problems of cold dark matter cosmology.

27.2 Dark Energy and the Cosmological Constant

The cosmological constant problem—why the observed vacuum energy is ~ 120 orders of magnitude smaller than naive quantum field theory estimates—is reframed in LPST. The vacuum energy is the energy of the maximally coherent substrate configuration, which is exactly zero at tree level (the Mexican-hat potential vanishes at its minimum). Quantum corrections generate a non-zero vacuum energy, but the relevant energy scale is set by the substrate parameters (α, A_0, ξ) , which are self-consistently determined. The smallness of Λ may reflect a near-cancellation between contributions from different topological sectors of the substrate, analogous to the cancellation mechanism in unimodular gravity [43].

Alternatively, dark energy could be a slowly evolving cosmological-scale phase gradient—a quintessence-like field that is simply the ultra-long-wavelength sector of the phase field φ . The equation of state of this “dark phase energy” is $w = p/\rho \approx -1 + \varepsilon$, where ε depends on the rate of change of the phase gradient on cosmological scales.

28. Relationship to Other Approaches

28.1 String Theory

Both LPST and string theory postulate a unified fundamental substrate (the complex field L vs. the string), but they differ in a critical respect: strings propagate in a pre-existing background spacetime (at least perturbatively), while LPST derives spacetime from the substrate. The worldsheet sigma model of string theory, which describes the string’s propagation in a target space, has structural similarities to the LPST polarization sector, where $\mathbf{u}$ maps from the emergent spacetime into the target manifold M_{int} . The topological aspects of string theory (D-branes, string junctions) have LPST counterparts in the various topological defects of the substrate. However, LPST does not require extra dimensions, supersymmetry, or a landscape of vacua.

28.2 Loop Quantum Gravity

LQG and LPST share the aspiration of background independence, but their implementations differ. LQG discretizes geometry at the Planck scale (spin networks, spin foams) [11,12], while LPST derives continuous geometry from a continuous field. The area and volume operators of LQG, which have discrete spectra, may have LPST counterparts in the discrete topological charges of the substrate. However, in LPST the discreteness of physical quantities is topological in origin (winding numbers, not operator spectra), and the continuous field description is maintained at all scales.

28.3 Causal Set Theory

Causal set theory [14] posits that spacetime is fundamentally a discrete partial order (a set of events with a binary causal relation). The metric is recovered in the

continuum limit from the number of causal-set elements in a given region (volume) and their causal relations. LPST shares the relational philosophy but uses correlation functions rather than partial orders, and maintains a continuous substrate.

28.4 Emergent Gravity Programs

LPST shares the philosophy of Verlinde [6], Jacobson [5], and Padmanabhan [7] that gravity is not fundamental. The key difference is that LPST provides an explicit microscopic substrate and a concrete mechanism (phase stiffness modulation) for the emergence of gravity, whereas the thermodynamic/entropic approaches remain agnostic about the microscopic degrees of freedom.

28.5 Twistor Theory

Penrose’s twistors [17] are intimately connected with null structures in Minkowski space. LPST’s emphasis on light-like phase relations is closely aligned with the twistor philosophy. The twistor correspondence maps solutions of the massless field equations to cohomology classes on twistor space; the LPST analogue would be a correspondence between substrate configurations and geometric structures, which deserves further exploration.

28.6 AdS/CFT and Holography

LPST’s correlation-to-geometry map (Section 9) is analogous to the bulk-boundary dictionary of AdS/CFT. The correlation kernel $C(x,y)$ plays a role analogous to the boundary CFT two-point function, and the emergent metric is analogous to the reconstructed bulk geometry. However, LPST does not require an asymptotically AdS boundary, and the “holography” is built into the substrate itself rather than being a duality between two different theories. This suggests that LPST may provide a more general framework in which holographic dualities are special cases.

29. Numerical Methods and Simulation Strategies

29.1 Lattice Discretization

The most straightforward computational approach to LPST is to discretize the substrate field on a four-dimensional hypercubic lattice with lattice spacing $a \leq \xi$. At each lattice site, the field variables are $(A, \varphi, \mathbf{u}) \in \mathbb{R}_+ \times S^1 \times S^2$. The action (5) is discretized using standard finite-difference prescriptions for the derivatives. The emergent metric is computed from the lattice correlation function $C(x,y)$ at each self-consistency iteration.

29.2 Self-Consistent Iteration

The self-referential nature of LPST (the metric depends on correlations which depend on the metric) necessitates an iterative solver:

1. Start with an initial trial metric $g_{\mu\nu}^{(0)}$ (e.g., flat Minkowski).
2. Solve the substrate field equations on this metric to obtain $L(x)$.
3. Compute the correlation kernel $C(x,y)$ from the solution.
4. Extract the emergent metric $g_{\mu\nu}^{(1)}$ from $C(x,y)$ via Eq. (24).
5. Compare $g_{\mu\nu}^{(1)}$ with $g_{\mu\nu}^{(0)}$. If they agree to within a specified tolerance, the solution is self-consistent. Otherwise, use $g_{\mu\nu}^{(1)}$ (or a weighted average) as the new trial metric and return to step 2.

Convergence of this iteration is not guaranteed in general but is expected for configurations close to the vacuum and for perturbative deviations. A proof of convergence in the full nonlinear regime is an important open mathematical problem.

29.3 Monte Carlo Methods

The path integral (49) can be evaluated numerically using Monte Carlo sampling of phase-consistent substrate configurations. This is analogous to lattice QCD Monte Carlo, but with the added complication of the self-consistent metric. Importance

sampling using the Boltzmann weight $\exp(-S[L])$ (in Euclidean signature) can be implemented using standard Metropolis or hybrid Monte Carlo algorithms.

29.4 Proposed Benchmark Simulations

We propose three benchmark simulations to validate LPST numerically:

6. **Single vortex defect:** Form a single phase vortex ($n = 1$) on the lattice, compute its energy profile, and measure the emergent gravitational field from the curvature of the correlation-derived metric. Compare to the Schwarzschild solution.
7. **Hopfion scattering:** Initialize two hopfions ($H = 1$) with opposite momenta and simulate their collision. Track the topological charges through the interaction and measure the scattering cross-section.
8. **Cosmological phase ordering:** Starting from a random (maximally incoherent) initial configuration, simulate the evolution of the substrate and track the growth of the correlation length $\xi(t)$ and the emergent scale factor $a(t)$. Compare to the Friedmann equation predictions.

30. Experimental and Observational Signatures

LPST generates several falsifiable predictions that distinguish it from standard general relativity and quantum field theory.

30.1 Vacuum Birefringence-like Effects

In regions of extreme phase-texture gradients (near compact objects, in the early universe), the emergent metric for different polarization modes of the phase field may differ, producing a birefringence-like effect: the two polarization states of light propagate at slightly different speeds. This is testable with next-generation X-ray polarimetry missions, building on the capabilities demonstrated by IXPE.

30.2 Discrete Defect Resonance Spectra

The mass spectrum of topological defects in LPST is determined by the topology of the hopfion and skyrmion solution spaces. Specific mass ratios are predicted by the topological invariants, and these ratios are independent of the substrate parameters (they depend only on the topology of M_{int}). If LPST is correct, the mass ratios of certain elementary particles should match the predicted ratios—a highly non-trivial test.

30.3 Lorentz Invariance Violation

As discussed in Section 12.3, LPST predicts energy-dependent speed-of-light deviations at ultra-high energies. Current Fermi-LAT constraints [32] are consistent with LPST for $n \geq 2$. The Cherenkov Telescope Array (CTA) will probe the relevant parameter space for quadratic LIV.

30.4 Gravitational Wave Echoes

The correlation-defined horizon of a black hole defect has a finite width $\sim \xi$, rather than being an infinitely thin surface. This may produce echoes in the gravitational wave ringdown signal following a black hole merger: gravitational waves partially reflected from the inner boundary of the horizon arrive at the detector delayed by $\sim \xi/c$ relative to the main signal. Current LIGO/Virgo/KAGRA data have placed tentative (though not yet definitive) constraints on echo signals [42].

30.5 Modified Strong-Field Gravitational Waves

In the strong-field regime (binary black hole mergers, neutron star mergers), the higher-derivative corrections to the effective gravitational action (Eq. 48) modify the gravitational waveform at high frequencies. These deviations are parameterized by the coefficients c_1, c_2 , which are of order $\xi^2 \sim l_P^2$. While extremely small, next-generation detectors (LISA, Einstein Telescope, Cosmic Explorer) may achieve the sensitivity needed to detect or constrain these effects.

30.6 Anomalous Magnetic Moments

Topological self-interaction corrections to the defect's electromagnetic coupling may modify the anomalous magnetic moments of leptons. The corrections are of order $(\xi/r_{\text{defect}})^2$, where r_{defect} is the defect size. For the electron ($r_{\text{defect}} \sim \lambda_C \sim 10^{-12}$ m) and $\xi \sim l_p \sim 10^{-35}$ m, the correction is $\sim 10^{-46}$ —far below current experimental sensitivity, but a prediction in principle.

30.7 Non-Gaussian CMB Signatures

The topological phase-ordering epoch that drives inflation in LPST may produce distinctive non-Gaussian features in the CMB temperature and polarization power spectra, arising from the non-linear dynamics of topological defect formation and annihilation during the ordering process.

30.8 Dark Matter Signatures

If dark matter consists of diffuse topological defect configurations, their spatial distribution may exhibit characteristic correlations that differ from standard WIMP or axion predictions. In particular, the minimum-size constraint on defect configurations may produce a distinctive core profile in dark matter halos, testable with high-resolution gravitational lensing surveys.

31. Limitations and Open Problems

LPST is presented as a framework and a research program, not as a finished theory. Several critical issues remain unresolved.

31.1 Standard Model Particle Spectrum

While LPST provides natural mechanisms for gauge symmetries (Section 16) and topological quantum numbers (Section 14), the specific identification of all Standard Model particles with particular defect configurations is incomplete. Reproducing the observed mass spectrum of quarks and leptons from the parameters α , β , γ , and λ is a major computational and theoretical challenge.

31.2 Renormalization Behavior

The ultraviolet behavior of defect interactions and the renormalizability (or non-perturbative UV completeness) of the substrate theory are undeveloped. The self-referential nature of the theory (the metric depends on the field, which depends on the metric) may introduce novel UV/IR mixing effects that require new mathematical tools to analyze.

31.3 Fermion Generations

The Standard Model contains three generations of quarks and leptons. In LPST, these should correspond to different topological sectors or excitations of the defect configurations, but the mechanism that produces exactly three generations has not been derived. It may be related to the homotopy groups of M_{int} or to the stability properties of higher-charge defects.

31.4 CP Violation

CP violation in the Standard Model arises from the complex phase of the CKM matrix. In LPST, this must correspond to a topological or geometric property of the defect interactions. The identification of this mechanism is an open problem.

31.5 Neutrino Masses

The extreme smallness of neutrino masses (at least six orders of magnitude smaller than the electron mass) requires a topological explanation. One possibility is that neutrinos are defect configurations with a nearly exact topological symmetry that is weakly broken, generating a small but non-zero mass via a see-saw-like mechanism in the defect sector.

31.6 The Hierarchy Problem

The large disparity between the electroweak scale ($\sim 10^2$ GeV) and the Planck scale ($\sim 10^{19}$ GeV) is the hierarchy problem. LPST may reframe this problem: if both scales are determined by substrate parameters (α , A_0 , ξ), the hierarchy may correspond to a natural separation between the energy scale of the smallest topological defects and the energy scale at which the substrate description itself breaks down.

31.7 Experimental Probes

Many of the predicted effects (Section 30) are extremely small, requiring advances in ultra-high-coherence interferometry, X-ray polarimetry, gravitational wave detection, and CMB observation. The development of concrete experimental proposals with quantitative predictions for specific instruments is an important near-term goal.

31.8 Mathematical Rigor

Several derivations in this paper are formal and heuristic, relying on plausible physical reasoning rather than rigorous functional analysis. A mathematically rigorous formulation of the correlation-to-metric map, the phase consistency principle, and the self-consistent iteration procedure is needed. This may require developments in functional analysis, geometric measure theory, and topological field theory.

32. Conclusion

We have presented Light-Phase Substrate Theory (LPST), a comprehensive framework in which light-like phase interference is the sole fundamental entity, and all familiar structures of physics—space, time, matter, energy, and gravity—emerge from the topology and correlations of a single universal complex field. By eliminating background spacetime and replacing it with relational interference geometry, LPST offers a unified conceptual foundation capable of accommodating quantum phenomena, gravity, and matter within a single substrate.

The three key achievements of LPST as presented in this paper are:

9. **Derivation of spacetime geometry from correlations.** The emergent metric $g_{\mu\nu}$ is extracted from the two-point correlation kernel $C(x,y)$ of the substrate, providing a concrete realization of the long-standing intuition that geometry is entanglement/correlation structure. Causal structure, light cones, and the Lorentz group all emerge from this construction.

10. Identification of matter with topological solitons. Particles are knots, vortices, and braids of the substrate fields, stabilized by topology and classified by the homotopy groups of the target space. Mass is trapped phase energy, charge is winding number, and spin statistics follow from the topology of the configuration space.

11. Emergence of gravity as self-induced refraction. Massive defects modulate the local phase stiffness, creating an effective refractive index that deflects passing phase waves. All gravitational phenomena—geodesic bending, time dilation, gravitational redshift, the Schwarzschild and Kerr metrics—are refractive effects in the substrate. The Einstein equations are recovered as the self-consistency condition for the correlation-derived metric.

The path forward involves three interleaved efforts: *mathematical refinement* of the foundations (rigorous correlation-to-metric map, proof of self-consistent convergence, analysis of renormalization), *numerical simulation* of the substrate on lattice systems (benchmark tests, defect spectrum computation, cosmological ordering), and *experimental proposal development* (quantitative predictions for LIV, gravitational wave echoes, CMB non-Gaussianity, dark matter distributions). We believe that LPST provides a fertile and physically motivated framework for pursuing these goals, and we invite the community to engage with, critique, and develop the ideas presented here.

Appendix A: Detailed Derivation of the Emergent Metric

We present the full derivation of the emergent metric $g_{\mu\nu}$ from the correlation kernel $C(x,y)$.

Step 1: Definition. The correlation kernel is $C(x,y) = \langle L(x) L^*(y) \rangle$, where the expectation value is taken over the ensemble of phase-consistent configurations (or, equivalently, via the path integral).

Step 2: Emergent distance. Define $d^2(x,y) = -\ell^2 \ln(|C(x,y)|/|C(x,x)|)$. This satisfies $d(x,x) = 0$ and $d(x,y) \geq 0$.

Step 3: Taylor expansion. For $y = x + \delta x$ with $|\delta x|$ small, expand $C(x, x + \delta x)$ to second order. Writing $C(x,y) = |C(x,y)| e^{i\Theta(x,y)}$, we have:

$$\ln|C(x, x+\delta x)| = \ln|C(x,x)| + \partial_\mu \ln|C(x,y)||_{y=x} \delta x^\mu + \frac{1}{2} \partial_\mu \partial_\nu \ln|C(x,y)||_{y=x} \delta x^\mu \delta x^\nu + O(\delta x^3) \quad (A1)$$

Step 4: Vanishing of the linear term. For a Hermitian correlation kernel, $C(x,y) = C(y,x)^*$, so $|C(x,y)| = |C(y,x)|$. The function $f(y) = \ln|C(x,y)|$ has a maximum at $y = x$ (since correlations are maximal at zero separation), so $\partial_\mu f|_{y=x} = 0$. The linear term vanishes.

Step 5: Identification of the metric. From Steps 2-4:

$$d^2(x, x+\delta x) = -\ell^2 \cdot \frac{1}{2} H_{\mu\nu}(x) \delta x^\mu \delta x^\nu \quad (A2)$$

where $H_{\mu\nu}(x) = \partial_\mu \partial_\nu \ln|C(x,y)||_{y=x}$ is the Hessian of $\ln|C|$ at coincident points. Since $d^2 \geq 0$ and the Hessian is negative semi-definite (f has a maximum at $y = x$), the matrix $-H_{\mu\nu}$ is positive semi-definite. We identify:

$$g_{\mu\nu}(x) = -(\ell^2/2) H_{\mu\nu}(x) \quad (A3)$$

Step 6: Vacuum verification. For $C(x,y) = A_0^2 \exp(-\eta_{\alpha\beta} \Delta x^\alpha \Delta x^\beta / (2\xi^2))$, we compute $H_{\mu\nu} = -\eta_{\mu\nu}/\xi^2$, giving $g_{\mu\nu} = (\ell^2/(2\xi^2)) \eta_{\mu\nu}$. With the identification $\ell^2/(2\xi^2) = 1$, we recover the flat Minkowski metric.

Conditions for validity. The Taylor expansion requires that δx is small compared to the scale over which C varies, i.e., $|\delta x| \ll \xi$. The metric is a well-defined smooth tensor field only if $H_{\mu\nu}$ is non-degenerate, which fails at defect cores where C has singularities. At these points, the geometric description breaks down and one must resort to the full substrate description.

Appendix B: Topological Classification of Defects

The field space of the substrate is $F = S^1 \times S^2$ (phase times polarization, for the minimal model with $M_{\text{int}} = S^2$). The relevant homotopy groups are:

Homotopy Group	Value	Physical Interpretation
$\pi_0(S^1 \times S^2)$	0	Connected; no domain walls
$\pi_1(S^1 \times S^2)$	$\mathbb{Z}$	Phase vortex lines (electric charge)
$\pi_2(S^1 \times S^2)$	$\mathbb{Z}$	Skyrmions / monopoles
$\pi_3(S^1 \times S^2)$	$\mathbb{Z} \oplus \mathbb{Z}$	Hopfions + phase instantons

Here $\pi_1(S^1 \times S^2) = \pi_1(S^1) \times \pi_1(S^2) = \mathbb{Z} \times 0 = \mathbb{Z}$, from the phase winding. $\pi_2(S^1 \times S^2) = \pi_2(S^1) \times \pi_2(S^2) = 0 \times \mathbb{Z} = \mathbb{Z}$, from the polarization wrapping. $\pi_3(S^1 \times S^2) = \pi_3(S^1) \times \pi_3(S^2) = 0 \times \mathbb{Z} = \mathbb{Z}$ (the Hopf fibration), but the product structure adds $\pi_3(S^1) = 0$, so $\pi_3(S^1 \times S^2) = \mathbb{Z}$ from the Hopf invariant of the S^2 factor.

Hopfion construction. The hopfion is a map $\mathbf{u}: \mathbb{R}^3 \cup \{\infty\} \cong S^3 \rightarrow S^2$ with unit Hopf invariant $H = 1$. Explicitly, using the Hopf map $\pi: S^3 \rightarrow S^2$ composed with the stereographic identification $\mathbb{R}^3 \cup \{\infty\} \cong S^3$, the hopfion field is:

$$\mathbf{u}(\mathbf{x}) = (2(x_1x_3 + x_2r), 2(x_2x_3 - x_1r), r^2 - x_3^2) / (r^2 + x_3^2) \quad (\text{B1})$$

where $r^2 = x_1^2 + x_2^2$. The preimages of any two distinct points on S^2 are linked circles in $\mathbb{R}^3$, with linking number equal to the Hopf invariant $H = 1$. The energy of this configuration, computed from the action (5), gives the hopfion mass in terms of the substrate parameters.

Appendix C: Recovery of the Schwarzschild Solution

Setup. We seek a static, spherically symmetric configuration of the substrate that is self-consistent: the metric derived from its correlations matches the metric used in the field equations.

Step 1: Ansatz. Assume the emergent metric takes the isotropic form $ds^2 = -B(r)dt^2 + C(r)(dr^2 + r^2 d\Omega^2)$, with B and C to be determined. Assume the substrate amplitude is $A(r) = A_0 f(r)$ with $f(\infty) = 1$ and the phase is $\varphi = \omega t$ (static defect viewed in the rest frame).

Step 2: Amplitude equation. In the weak-field limit, the amplitude equation (14) with the Mexican-hat potential yields, outside the defect core:

$$\nabla^2 f \approx -(\omega^2/c^2) f(1 - f^2) \quad (C1)$$

For a localized defect, the solution at large r is $f(r) \approx 1 - k/r + O(1/r^2)$, where k is proportional to the defect's mass M .

Step 3: Correlation kernel. For the static solution with $A(r) = A_0(1 - k/r)$, the correlation kernel between a point at radius r and a point at infinity is $C(r, \infty) \approx A_0^2(1 - k/r)$. Following the metric extraction procedure (Section 9), the emergent metric components are:

$$B(r) = 1 - 2k/r, \quad C(r) = 1 + 2k/r \quad (C2)$$

to leading order in k/r .

Step 4: Identification. Comparing with the linearized Schwarzschild metric, $ds^2 = -(1 - 2GM/(rc^2))c^2dt^2 + (1 + 2GM/(rc^2))(dr^2 + r^2d\Omega^2)$, we identify:

$$k = GM/c^2 \quad (C3)$$

where M is the total phase energy of the defect and $G = c^4/(8\pi\alpha A_0^2)$ is Newton's constant expressed in terms of substrate parameters.

Step 5: Self-consistency check. The metric (C2) is used in the field equations, which yield the amplitude profile $f(r) = 1 - k/r$, which produces the correlation kernel that yields the metric (C2). The solution is self-consistent to leading order in k/r . Higher-order corrections require iterating the self-consistency procedure and produce $O(k/r)^2$ corrections to the metric—these match the post-Newtonian corrections of the exact Schwarzschild solution.

Appendix D: Status and Research Directions

This document defines LPST as a complete foundational framework, suitable for: further mathematical refinement, numerical simulation, experimental proposal development.

D.1 Near-Term Goals (1-3 years)

- Rigorous formulation of the correlation-to-metric map, including existence and uniqueness theorems for the emergent metric under physically reasonable conditions on $C(x,y)$.
- Lattice simulation of single defect configurations (vortex, skyrmion, hopfion) with self-consistent metric iteration.
- Computation of the defect mass spectrum for the minimal model ($M_{\text{int}} = S^2$) and comparison with observed particle masses.
- Quantitative predictions for Lorentz invariance violation parameters, with comparison to existing Fermi-LAT and future CTA constraints.

D.2 Medium-Term Goals (3-7 years)

- Identification of M_{int} that reproduces the Standard Model gauge group $SU(3) \times SU(2) \times U(1)$.
- Derivation of the three-generation structure from the topological classification of defects.
- Full numerical simulation of two-defect scattering with self-consistent gravity.
- Cosmological simulation of the phase-ordering epoch, including predictions for CMB non-Gaussianity.
- Analysis of renormalization group flow and UV/IR structure of the substrate theory.

D.3 Long-Term Goals (7+ years)

- Complete mathematical proof of the emergence of general relativity from the substrate self-consistency condition.
- Derivation of all Standard Model parameters (masses, coupling constants, mixing angles) from the substrate parameters.

- Experimental detection or exclusion of LPST-specific signatures (gravitational wave echoes, LIV, dark matter halo profiles).
 - Development of LPST-based quantum gravity phenomenology applicable to early universe, black hole, and high-energy collider physics.
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